UniPool

Share the Ride. Split the Fare.

Group 16 • CS253 Software Development and Operations • IIT Kanpur

Team : Undefined Behavior

GitHub Repo: https://github.com/naman065/UniPool

The Problem

Getting off campus is a pain — every single time.

Auto fares are expensive

Solo trips from IITK cost ₹150–₹200. Nobody wants to pay alone.

WhatsApp groups get spammed

Students flood batch chats just to find one co-rider.

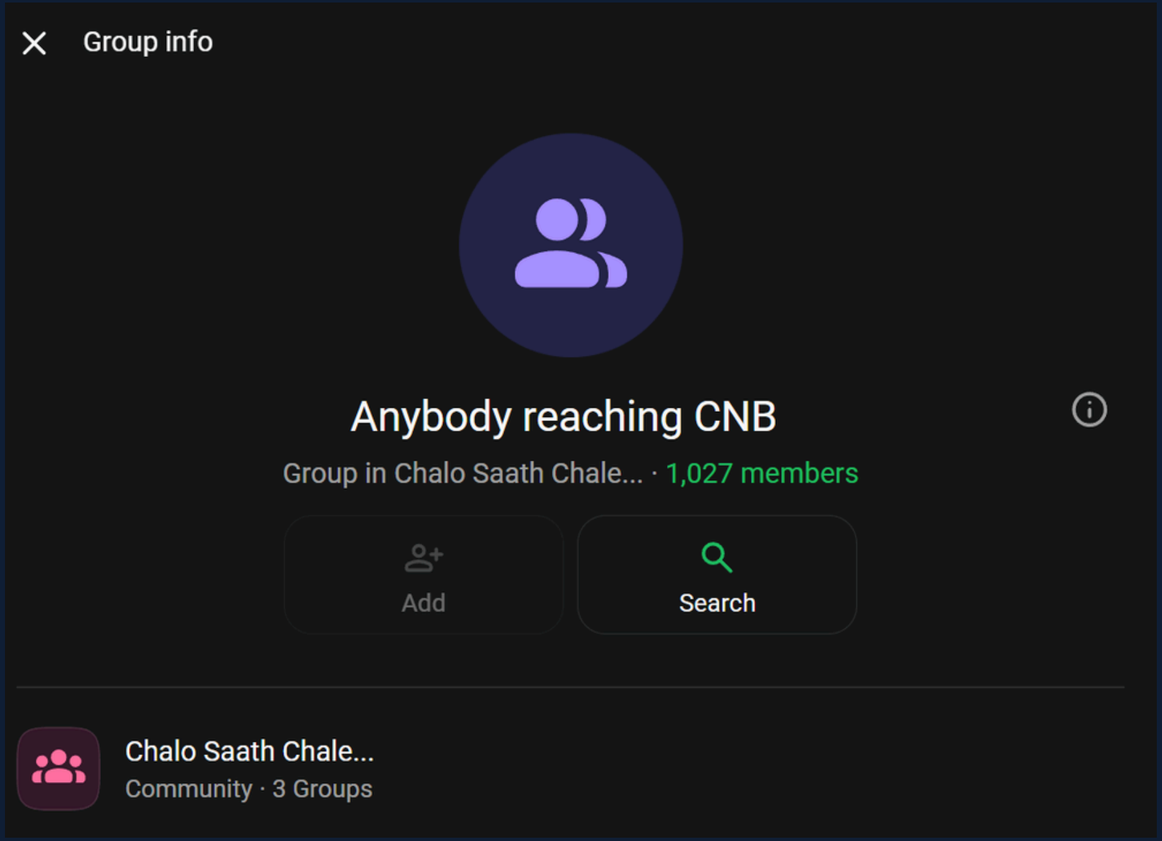
No way to discover co-riders

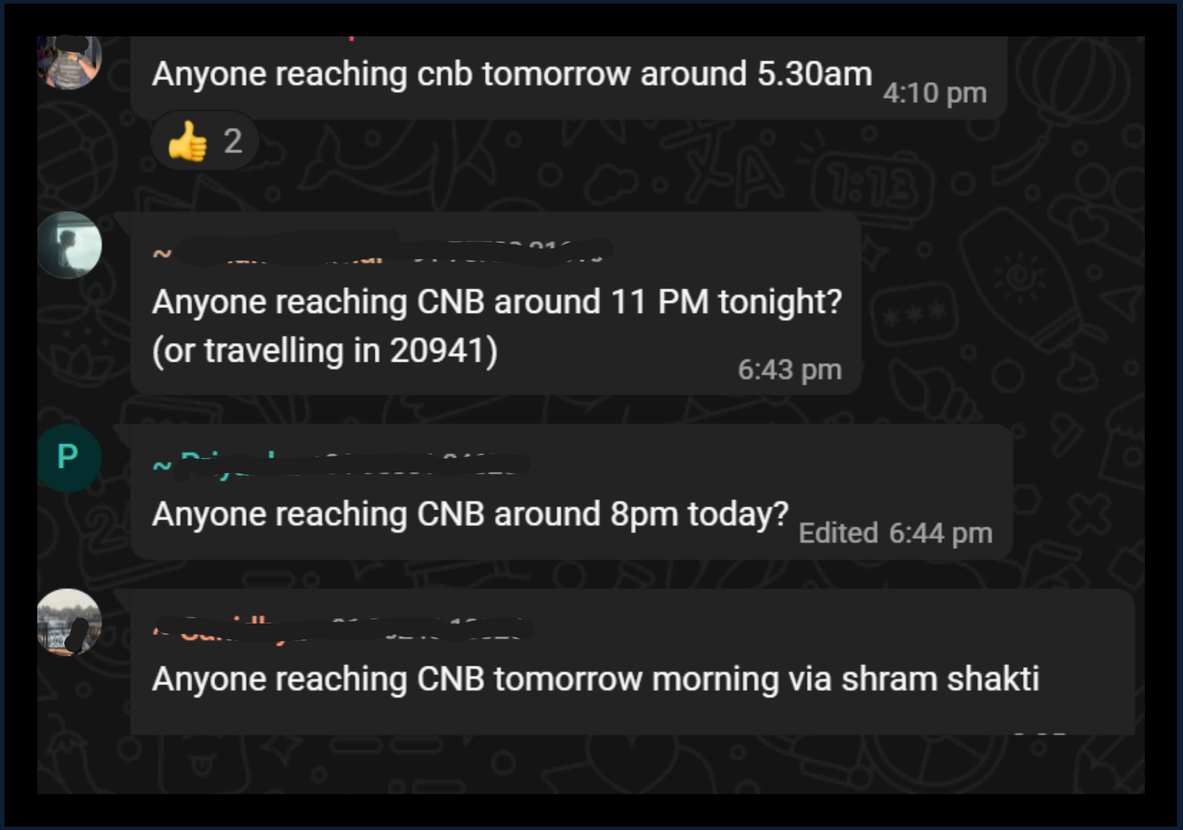
No central place to see who's going your way on the same day.

Coordination is chaotic

After finding someone, pickup timing still happens across 3 different chats.

The Problem


Group info
Anybody reaching CNB
Group in Chalo Saath Chale... · 1,027 members
Add
Search
Chalo Saath Chale...
Community · 3 Groups


Anyone reaching cnb tomorrow around 5.30am 4:10 pm
2
Anyone reaching CNB around 11 PM tonight?
(or travelling in 20941) 6:43 pm
P
Anyone reaching CNB around 8pm today? Edited 6:44 pm
Anyone reaching CNB tomorrow morning via shram shakti

Our Solution — UniPool

One app. No more WhatsApp chaos.

Post a Ride

Leaders set route + date. Instantly broadcasted to all user of the app

Find & Join

Browse open rides, filter by destination and time of the day, join with one tap.

Chat Inside

Each ride has its own chat thread — coordinate and plan the ride.

Tech Stack & Architecture

Frontend

- Flutter/Dart (Native cross platform compilation)
- Strongly-Typed State Management
- Polymorphic Custom Widgets
- Material Design 3 Theming Engine

Backend & DB

- Google Cloud Serverless infrastructure
- Firestore NoSQL Document Tree
- Real-Time Notification & Chat Websockets
- Instant offline support(Caching of backend data until reconnect)

App Architecture

- Decouple Repository Pattern
- Atomic Transactions
- Persistent StreamBuilders
- Strict Object-Oriented Models

Core Requirements — Post & Find Rides

Lead a Ride

- Select route (IITK/Kanpur) & travel date → publish ride
- Manage ride (edit/delete, notify participants)
- Complete ride → participants rate
- Completed rides tracked on leader profile

Find a Ride

- Filter rides (destination & time) and explore options
- Check leader profile, participants & ratings
- Request to join ride
- Track all rides in “My Rides”

Core Requirements — Chat & Security

Real-Time Group Chat

- Per-ride dedicated message thread
- Used for pickup time, gate location, last-min updates
- Full message history persists on app reopen

Auth & Security

- Email/password login with mandatory email verification
- Password reset via Firebase email link

DEMO TIME

Future Development Plans

01

Google Maps Support

- Currently there are preset location to choose from, for source and destination
- We wish to introduce more locations for diverse campus needs

02

In-App Fare Splitting

- Auto-calculate per-person cost for common routes
- Track who has paid inside the ride thread
- UPI deep-link integration for direct payment

03

Push Notifications

- Currently we have in-app notifications for updates regarding rides that a rider has wished to join
- We use servers free of cost and this kind of service requires paid server
- We could also use this to give reminders about the ride

Lessons Learned

Git Workflow Matters

Merge conflicts taught us to branch early and open PRs before merging to main.

Plan Driven Process

Software Development is not only implementation and bug fixing but also proper documentation

Institutional Firewall Issue

Campus SMTP blocked email delivery — Firebase's own verification links were our workaround.

Debugging Effectively

Learnt how to actually debug without solely relying on LLMs and stress testing ourselves

Team Work and Collaboration

Learned to work on a 3 month long project as a team with managing deadlines and division of work.

QnA

Acknowledgements

Prof. Indranil Saha

For structured guidance and teaching CS253 — the project framework made us better engineers.

Mentor TA: V Arvind

For timely and detailed feedback on our SRS, design review, and implementation milestones.

Beta Testing Group: Group 17

For testing the app, catching real bugs, and giving honest feedback before final submission.

Thank You!

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